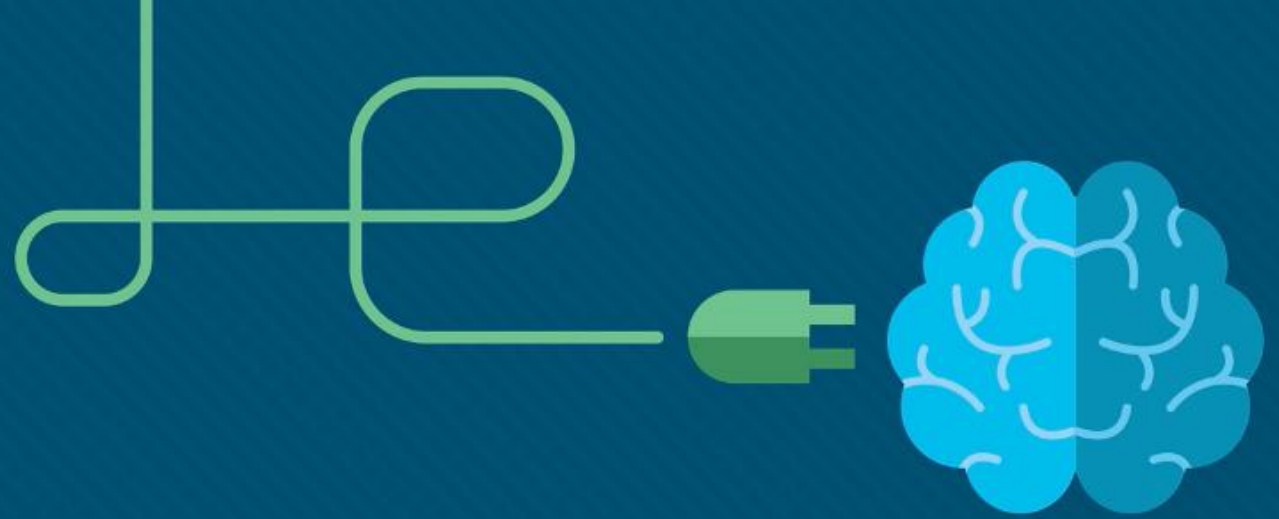


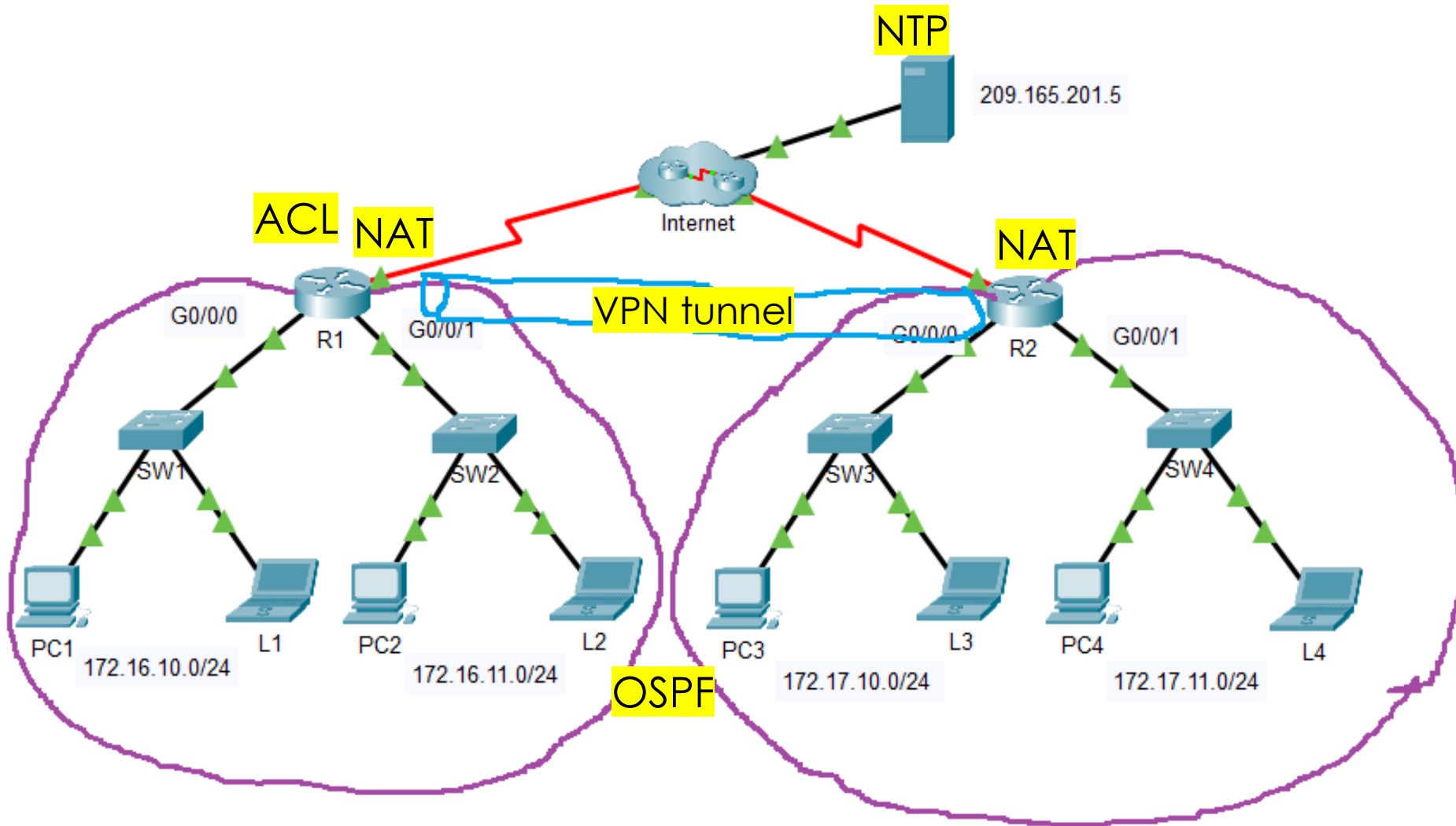


TSHOOT

Modul 12



Hvad kan jeg efter Netværk II?



Samt "kendskab til" Design; Management; Virtualisering;
Automatisering; QoS og WAN .



Meget kort kapital 😊

Ca. 20 dias



Overview

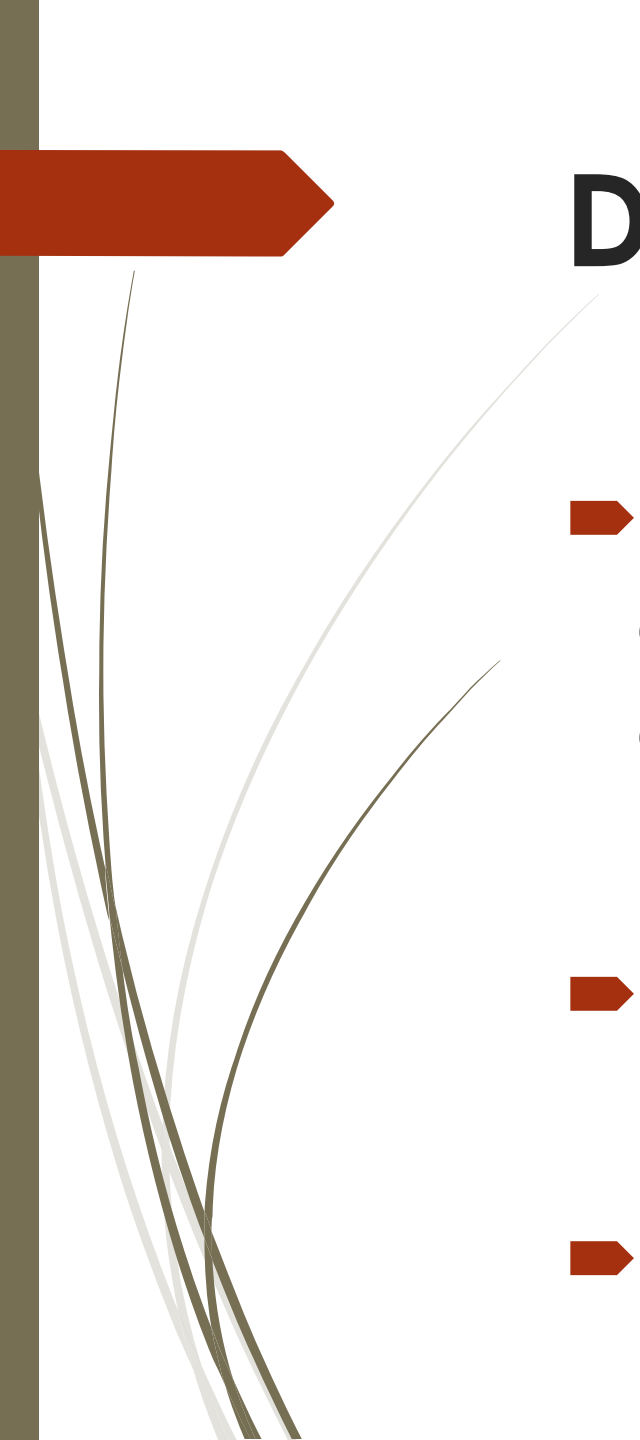
- Troubleshooting with a systematic approach
 - Network Troubleshooting
- 

Network Failure is BAD



FAILURE

A red, distressed stamp with the word "FAILURE" in bold, uppercase letters. The stamp is tilted and surrounded by red ink splatters and a small gear icon on the right side.

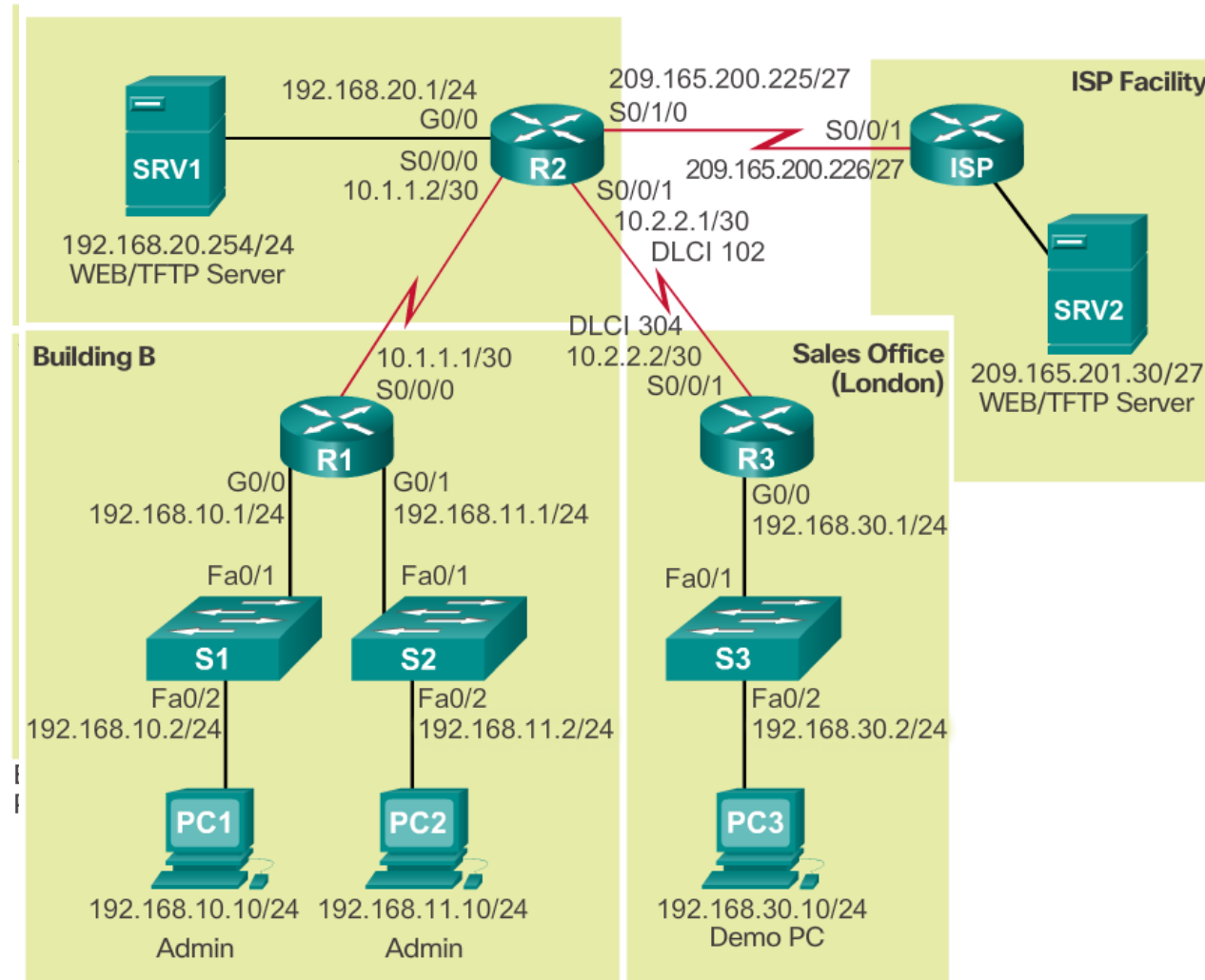


Documenting the network

- Configuration files, including network configuration files and end-system configuration files
- Physical and logical topology diagrams
- A baseline performance level

Network Topology Diagram

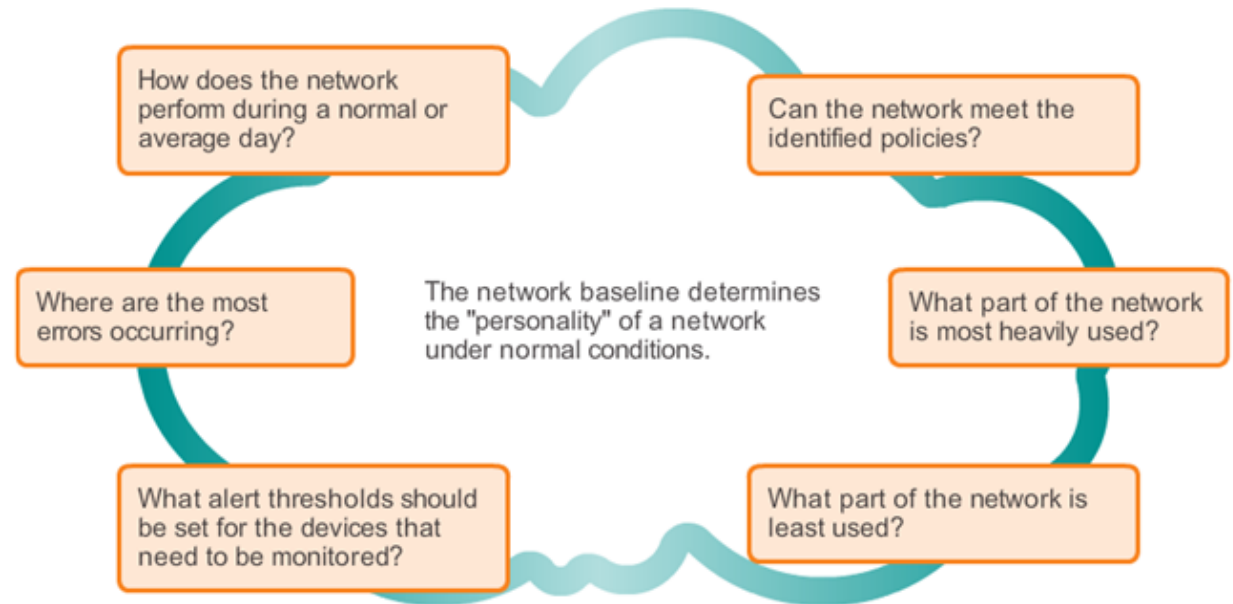
Logical IPv4 Network Topology



Network Baseline

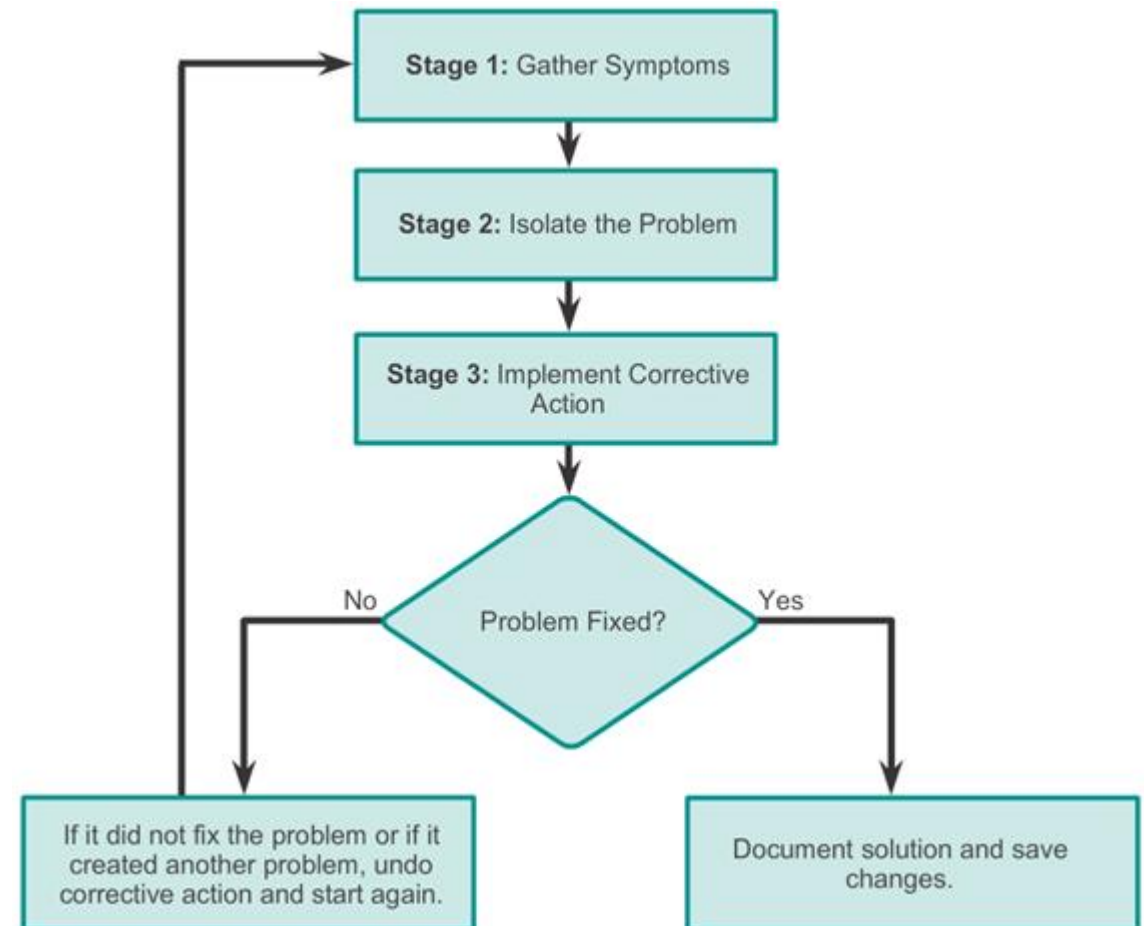
- **Step 1.**
Determine what types of data to collect.
- **Step 2.** Identify devices and ports of interest.
- **Step 3.**
Determine the baseline duration.

Questions that a Network Baseline Answers



General Troubleshooting Procedures

- Document unsuccessful solution too!
- Timestamp it



Gathering Symptoms

Alle disse kommandoer kan bruges til at opbygge din netværks dokumentation

- Commands that are useful for gathering symptoms
- ALSO very useful for documenting the network.

Command	Description
<code>ping {host ip-address}</code>	Sends an echo request packet to an address, then waits for a reply. The <i>host ip-address</i> variable is the IP alias or IP address of the target system.
<code>tracert {destination}</code>	Identifies the path a packet takes through the networks. The destination variable is the hostname or IP address of the target system.
<code>telnet {host ip-address}</code>	Connects to an IP address using the Telnet application.
<code>show ip interface brief</code> <code>show ipv6 interface brief</code>	Displays a summary of the status of all interfaces on a device.
<code>show ip route</code> <code>show ipv6 route</code>	Displays contents of currently running configuration file.
<code>show running-config</code>	Displays a list of options for enabling or disabling debugging events on a device.
<code>[no] debug ?</code>	Displays a list of options for enabling or disabling debugging events on a device.
<code>show protocols</code>	Displays the configured protocols and shows the global and interface-specific status of any configured Layer 3 protocol.

Questioning End Users

"der er intet er virker – det er PISSE irriterende – vi har faktisk ret travlt"

"det her er nok det vigtigste i hele verden"

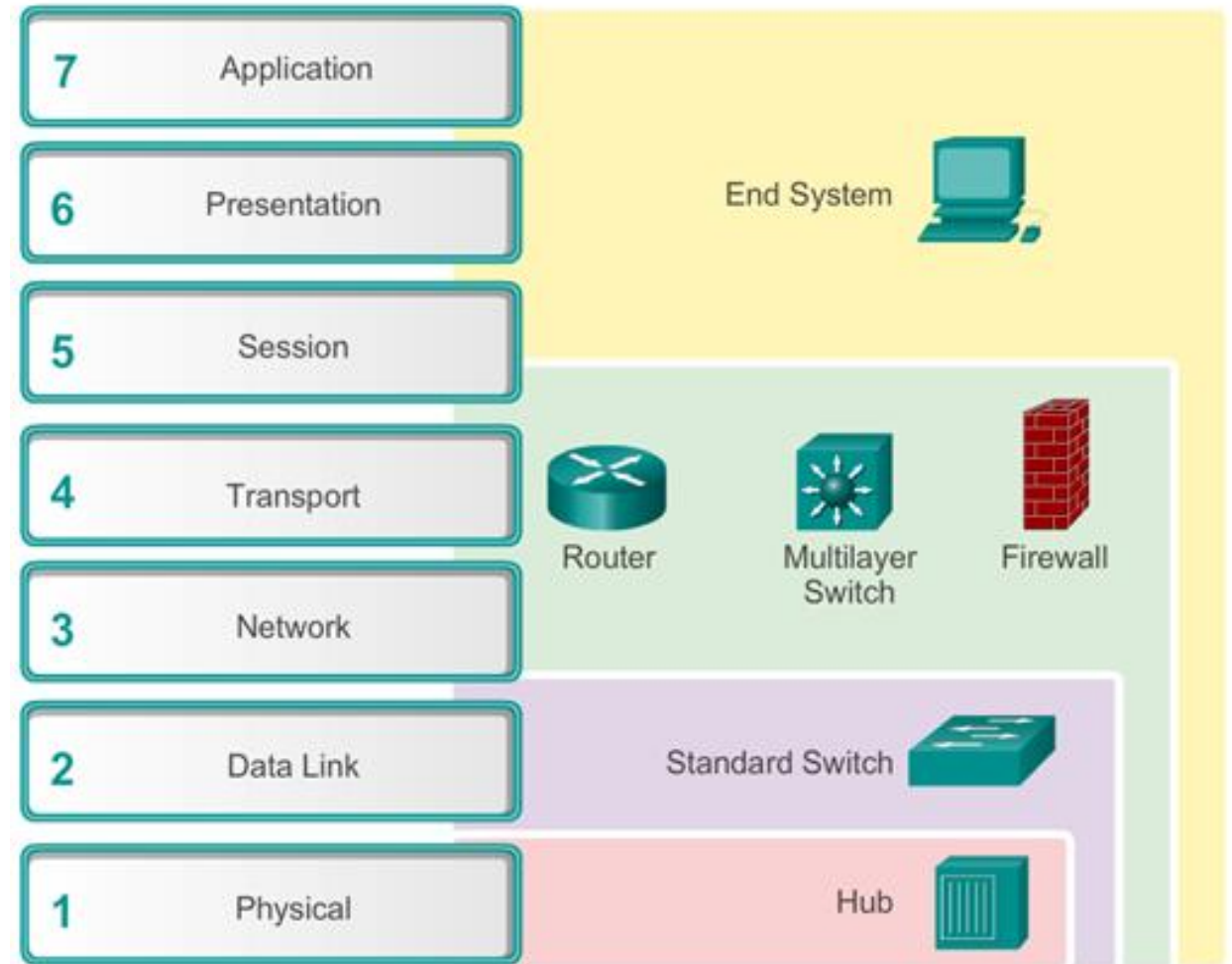
"virkede i hvert fald for 3 år siden"

Guidelines	Example End-user Questions
Ask questions that are pertinent to the problem.	What does not work?
Use each question as a means to either eliminate or discover possible problems.	Are the things that do work and the things that do not work related?
Speak at a technical level that the user can understand.	Has the thing that does not work ever worked?
Ask the user when the problem was first noticed.	When was the problem first noticed?
Did anything unusual happen since the last time it worked?	What has changed since the last time it did work?
Ask the user to recreate the problem, if possible.	Can you reproduce the problem?
Determine the sequence of events that took place before the problem happened.	When exactly does the problem occur?

Isolating the Issue Using Layered Models

- Compare characteristics of the problem to the layers of a logical network model
- Using the layered models, there are three primary methods for troubleshooting networks:
 - Bottom-up
 - Top-down
 - Divide-and-conquer

OSI Reference Model





Less Structured Approaches

In addition to the systematic, layered approach to troubleshooting, there are also, less-structured troubleshooting approaches:

- One troubleshooting approach is based on an **educated guess** by the network administrator, based on the symptoms of the problem.
- Another approach involves **comparing a working and nonworking situation**, and spotting significant differences.
- **Swapping the problematic device** with a known, working one is a quick way to troubleshoot.

Troubleshooting Tools

Common **software** troubleshooting tools include:

- NMS tools
- Knowledge bases
- Baselining tools
- Host-based protocol analyzers
- Cisco IOS EPC

Common **hardware** troubleshooting tools include:

- Network analysis module
- Digital multimeters
- Cable testers
- Cable analyzers
- Portable network analyzers

Troubleshooting with Syslog

	Level	Keyword	Description	Definition
Highest Level	0	emergencies	System is unusable	LOG_EMERG
	1	alerts	Immediate action is needed	LOG_ALERT
	2	critical	Critical conditions exist	LOG_CRIT
	3	errors	Error conditions exist	LOG_ERR
	4	warnings	Warning conditions exist	LOG_WARNING
	5	notifications	Normal but significant condition	LOG_NOTICE
	6	informational	Informational messages only	LOG_INFO
Lowest Level	7	debugging	Debugging messages	LOG_DEBUG

Troubleshooting Layer 1

7

Application

```
R1# show interfaces gigabitethernet 0/0
GigabitEthernet0/0 is up, line protocol is up
  Hardware is CN Gigabit Ethernet, address is
  d48c.b5ce.a0c0(bia d48c.b5ce.a0c0)
  Internet address is 10.1.10.1/24
  <output omitted>
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total
  output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    85 packets input, 7711 bytes, 0 no buffer
    Received 25 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 5 multicast, 0 pause input
    10112 packets output, 922864 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    11 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier, 0 pause output
    0 output buffer failures, 0 output buffers swapped out
R1#
```

Symptoms

- Performance lower than baseline
- Loss of connectivity
- Network bottlenecks or congestion
- High CPU utilization rates
- Console error messages

1

Physical

Causes

- Power-related
- Hardware faults
- Cabling faults
- Attenuation
- Noise
- Interface configuration errors

Troubleshooting Layer 2

Symptoms

- No functionality or connectivity at network layer or above
- Network operating below baseline performance levels
- Excessive broadcasts
- Console messages

```
Administrator: Kommandoprompt
C:\Windows\system32>arp -a

Interface: 10.193.1.138 --- 0xc
Internet Address      Physical Address      Type
10.128.64.1           00-17-94-5f-bb-41     dynamic
10.193.1.1            00-17-94-5f-bb-41     dynamic
10.193.1.2            f0-1f-af-6f-13-ea     dynamic
10.193.1.3            6a-05-ca-24-f9-24     dynamic
10.193.1.5            00-15-5d-35-7e-23     dynamic
10.193.1.6            b8-ca-3a-be-f3-cd     dynamic
10.193.1.7            d0-67-e5-35-2b-f1     dynamic
10.193.1.8            f0-1f-af-3b-d8-57     dynamic
10.193.1.9            f0-1f-af-54-5b-4a     dynamic
10.193.1.11           d0-67-e5-38-bc-8d     dynamic
10.193.1.12           00-25-22-99-e0-ef     dynamic
10.193.1.13           e0-db-55-de-c4-71     dynamic
10.193.1.16           98-90-96-c8-25-ee     dynamic
10.193.1.21           ec-f4-bb-53-6c-80     dynamic
10.193.1.22           d0-67-e5-54-e0-e6     dynamic
10.193.1.23           e0-db-55-de-c4-81     dynamic
10.193.1.24           d4-be-d9-75-64-6e     dynamic
10.193.1.26           f0-1f-af-23-7c-47     dynamic
10.193.1.255          ff-ff-ff-ff-ff-ff     static
10.193.16.200         00-17-94-5f-bb-41     dynamic
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.252           01-00-5e-00-00-fc     static
224.0.1.60            01-00-5e-00-01-3c     static
226.178.217.5         01-00-5e-32-d9-05     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Windows\system32>
```

Causes

- Encapsulation errors
- Address mapping errors
- Framing errors
- STP failure or loops

2

Data Link

1

Physical

Troubleshooting Layer 3

Verify the IPv6 Default Gateway

```
PC1> ipconfig
Windows IP Configuration
    Connection-specific DNS Suffix  : 
    IPv6 Address. . . . . : 
    2001:db8:acad:1:5075:d0ff:fe8e:9ad8
    Link-local IPv6 Address . . . . : fe80::5075:d0ff:fe8e:9ad8%13
    IPv4 Address. . . . . : 10.1.1.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1
                                10.1.10.1
```

Symptoms

- Network failure
- Suboptimal performance

4

Transport

3

Network

2

Data Link

Causes

- General network issues
- Connectivity issues
- Neighbor issues
- Topology database
- Routing table

Troubleshooting Layer 4

Common ACL Misconfigurations

- Selection of traffic flow
- Order of ACL entries
- Implicit deny all
- Address and IPv4 wildcard masks
- Selection of transport layer protocol
- Source and destination ports
- Use of the established keyword
- Uncommon protocols

Common Interoperability Areas

- BOOTP and DHCP
- DNS and WINS
- SNMP
- Tunneling and encryption protocols

NAT

7

Application

6

Presentation

5

Session

4

Transport

3

Network

2

Data Link

1

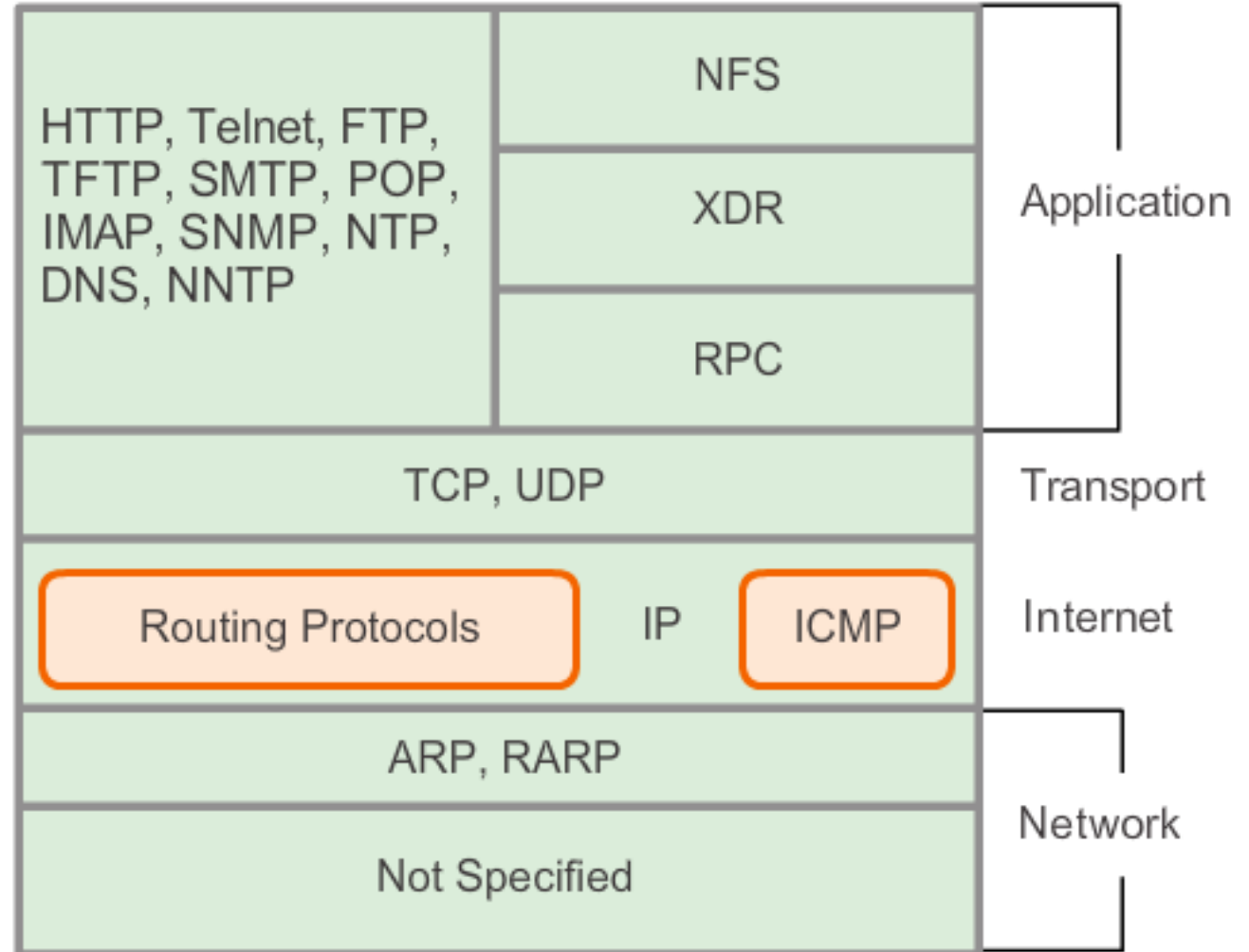
Physical

Troubleshooting Layer 5-7

OSI Reference Model

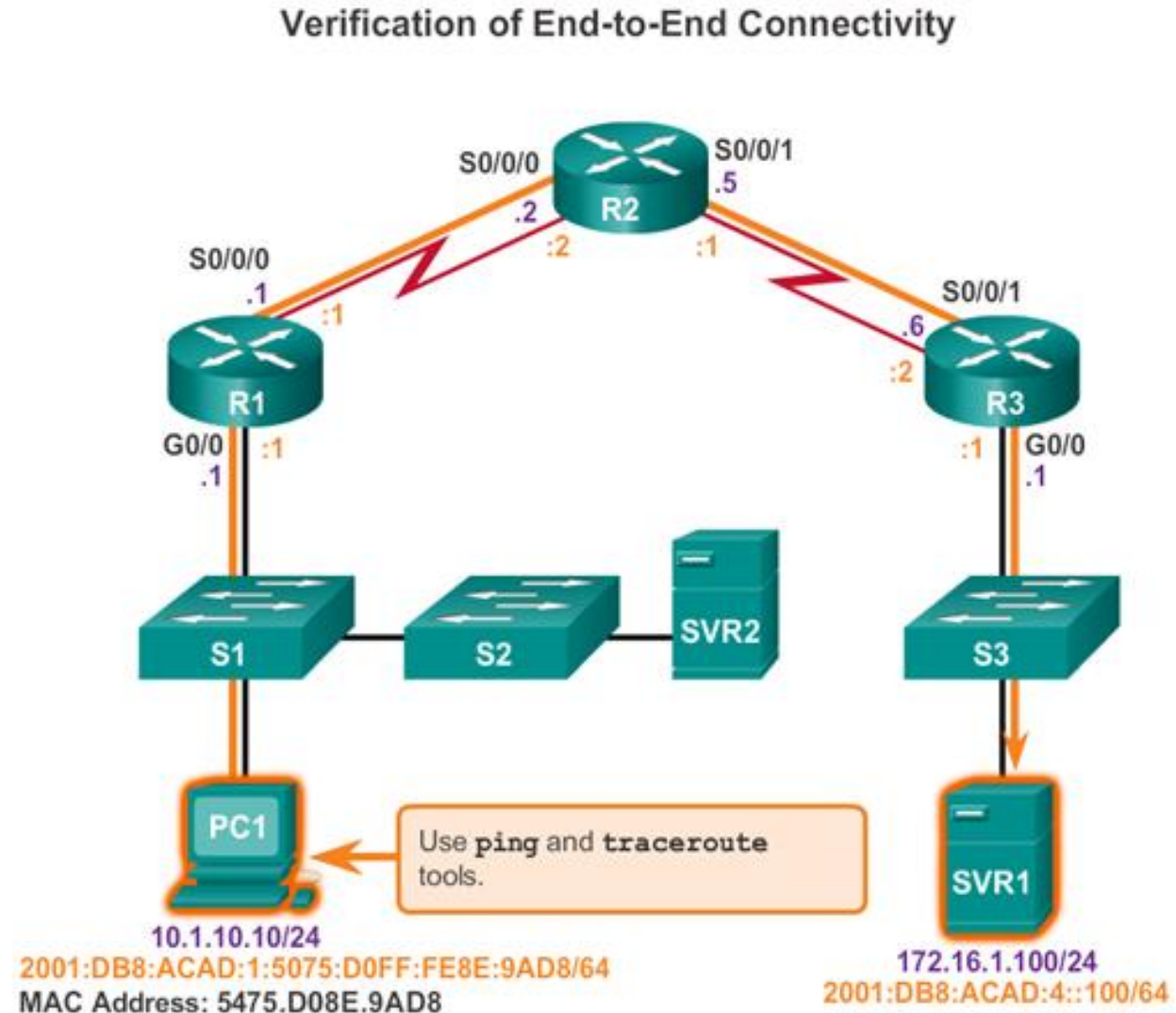


TCP/IP Reference Model



End-to-end connectivity testing

- A simple ping command can rule out ALOT of problems or indicate exactly what the problem is!





- Der er ikke nogen særskilt opgave til TSHOOT.
- I casen kommer SYSlog med.
- Læs Modul 12 og/eller se video 27+28